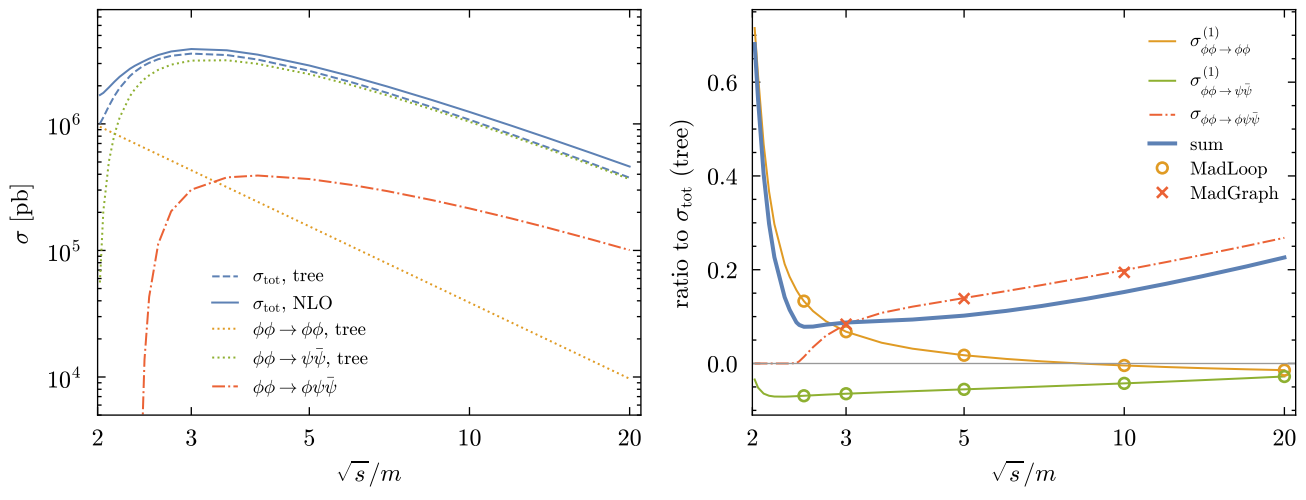
σ [pb]
10^6
10^5
10^4
$\sigma_{\rm tot}$, tree
$\sigma_{\rm tot}$, NLO
$\phi\phi \to \phi\phi$, tree
$\phi\phi \to \psi\bar{\psi}$, tree
$\phi\phi \to \phi\psi\bar{\psi}$
2
3
5
10
20
$\sqrt{s}/m$
ratio to $\sigma_{\rm tot}$ (tree)
0.6
0.4
0.2
0.0
$\sigma^{(1)}_{\phi\phi \to \phi\phi}$
$\sigma^{(1)}_{\phi\phi \to \psi\bar{\psi}}$
$\sigma_{\phi\phi \to \phi\psi\bar{\psi}}$
sum
MadLoop
MadGraph
2
3
5
10
20
$\sqrt{s}/m$